

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

OSCILLATIONS

LECTURE-04 **PHYSICS**

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Topic To Be Covered



(1) Relationship b/w V and x

$$\checkmark V = A\omega \cos(\omega t) \checkmark$$
$$V = \omega \sqrt{A^2 - x^2}$$

numerical.

(2) Damped oscillation

(3) Un damped oscillation

(4) Resonance

(5) Second pendulum.

(6) Periodic function

lecture no. 4

Relationship b/w (v) velocity and displacement (x)

As we know that

$$v = A\omega \cos \omega t$$

$$\therefore \sin^2 \omega t + \cos^2 \omega t = 1$$

$$\cos^2 \omega t = 1 - \sin^2 \omega t$$

On Squaring b/s

$$v^2 = (A\omega \cos \omega t)^2$$

$$v^2 = A^2 \omega^2 \cos^2 \omega t$$

$$v^2 = A^2 \omega^2 (1 - \sin^2 \omega t)$$

$$v^2 = \omega^2 (A^2 - A^2 \sin^2 \omega t)$$

→ 2025

→ 2024

→ 2021

$$v = \omega \sqrt{A^2 - x^2}$$

$$x = A \sin \omega t$$

$$v^2 = \omega^2 (A^2 - A^2 \sin^2 \omega t)$$

$$\underline{v^2} = \omega^2 (A^2 - \underline{x^2})$$

$$v = \sqrt{\omega^2 (A^2 - x^2)}$$

$$v = \sqrt{\omega^2} \sqrt{(A^2 - x^2)}$$

$$v = \omega \sqrt{A^2 - x^2}$$

numerical.

$$x = \underline{A \sin \omega t}$$

$$x^2 = (A \sin \omega t)^2$$

$$\underline{x^2} = \underline{A^2 \sin^2 \omega t}$$

$\omega \rightarrow$ angular velocity

$A \rightarrow$ Amplitude

$x \rightarrow$ displacement

Numerical.



$$\omega = 10 \text{ rad/sec}$$

$$A = 10\text{m} \quad | \quad 0.1\text{m}$$

$$x = 2\text{m} \quad | \quad 0.01\text{m}$$

$$V = ?$$

Sol

$$V = \omega \sqrt{A^2 - x^2}$$

$$V = 10 \sqrt{10^2 - 2^2}$$

$$V = 10 \sqrt{100 - 4}$$

$$V = 10 \sqrt{96}$$

$$V = 10 \times 9.77$$

$$V = 97.7 \text{ m/s.}$$

$$\sqrt{81} = 9 \quad 3$$

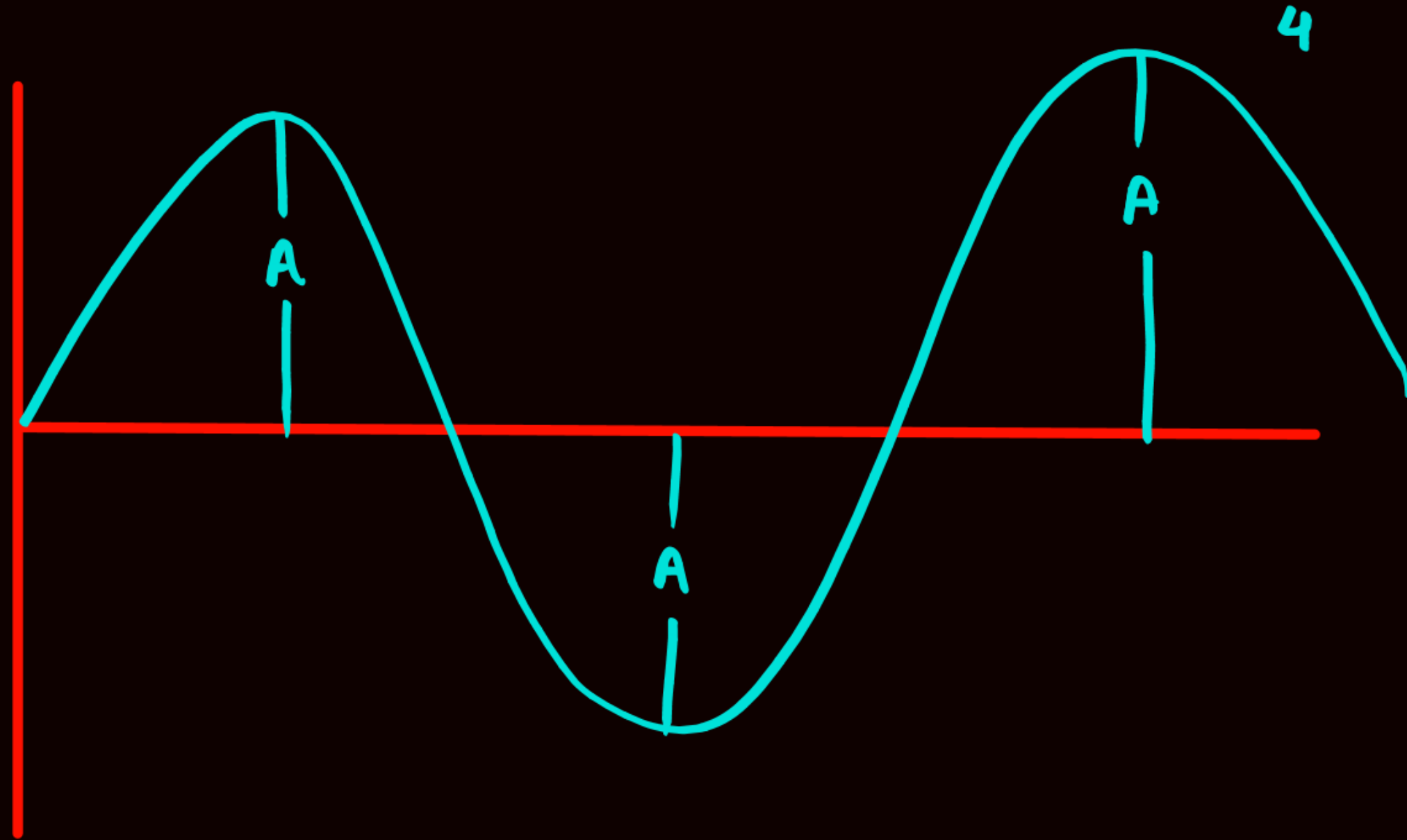
$$\sqrt{96} = 9.77$$

$$\sqrt{100} = 10$$

② UN DAMPED oscillation

- the oscillation which donot change its amplitude with time

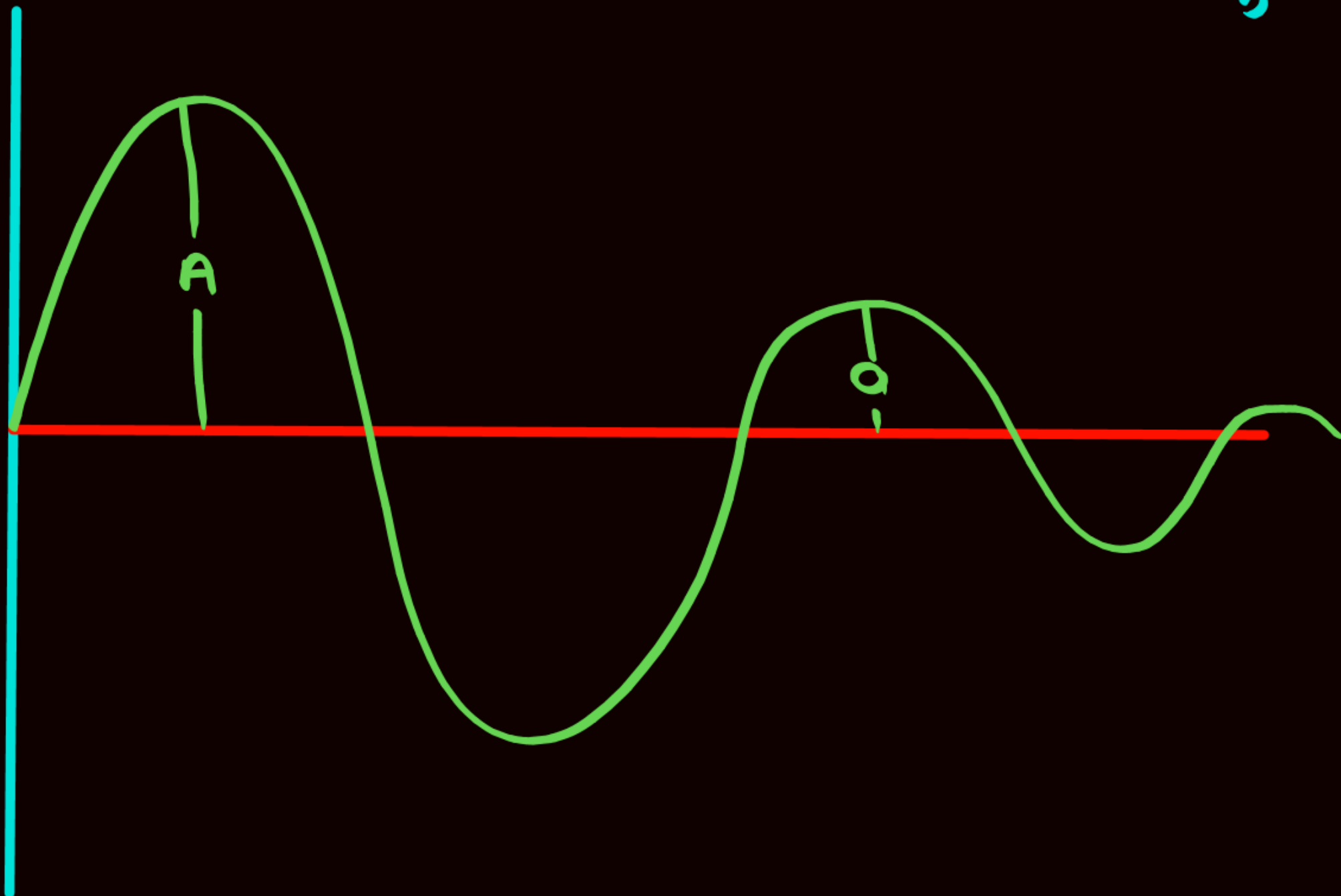
Amplitude = constant ✓



3

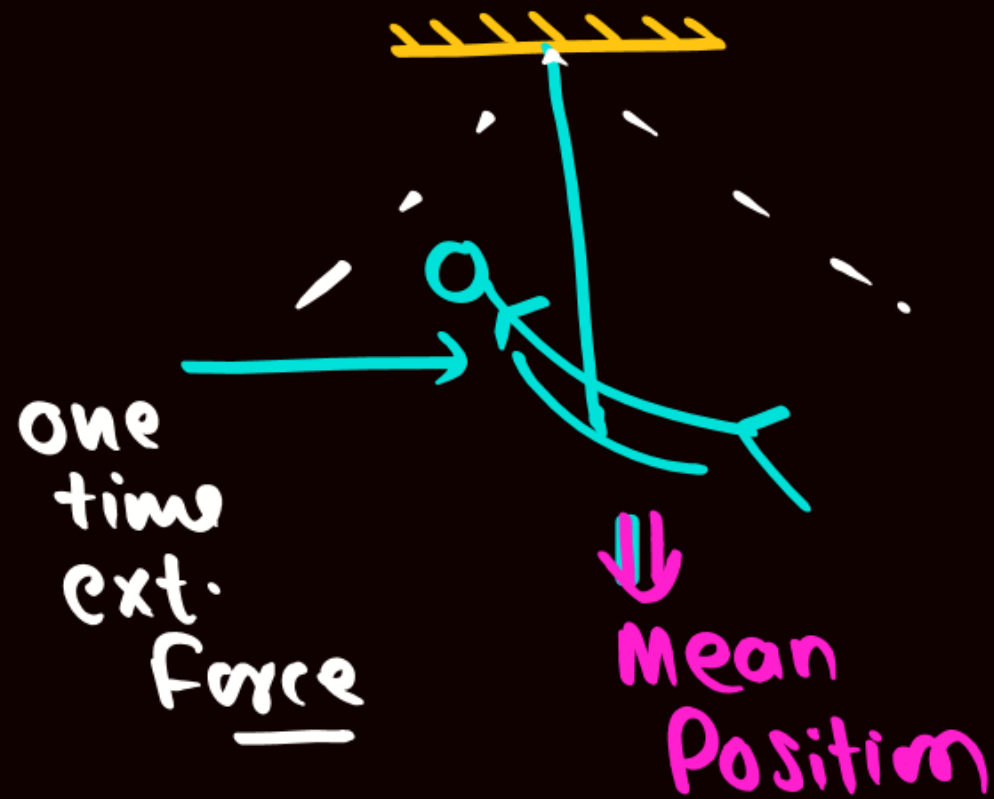
DAMPED Oscillation

when the Amplitude of
the oscillation
change with time



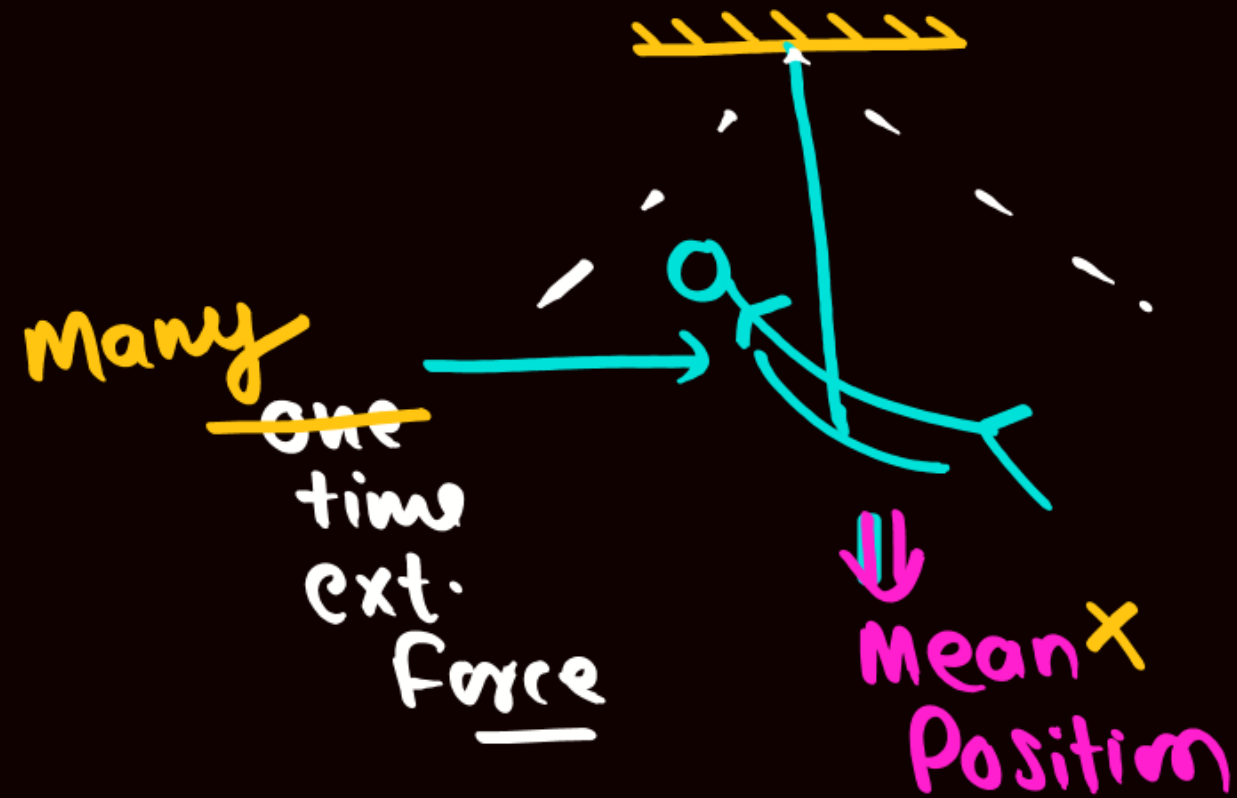
Free oscillation

when the particle oscillate with natural frequency



Force oscillation

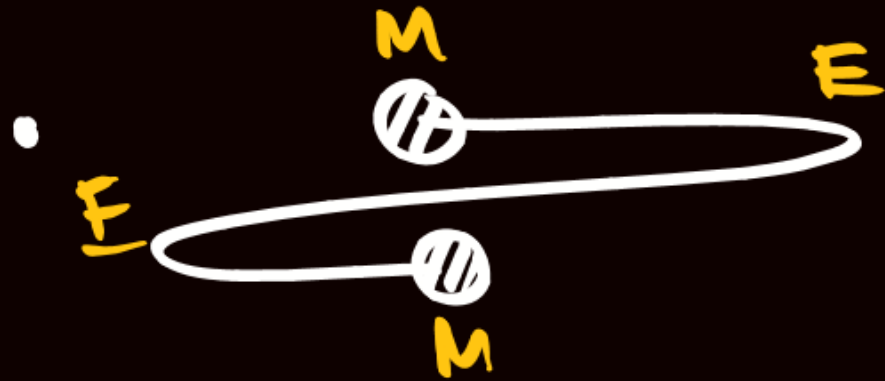
✓ when External Force is required to maintain the Amplitude of particle



⑥

✓ Frequency (ν) / f

• Repeat ✓



• the no. of oscillation
per unit time

$$f = \frac{n}{T}$$

$$f = \frac{n}{t} = \frac{1}{s} = s^{-1} = \underline{\text{Hertz}} \checkmark$$

✓ WM

$$\frac{1}{24}$$

ν_1

✓ HS

$$\frac{2}{24}$$

ν_2

✓ AS

$$\frac{4}{24}$$

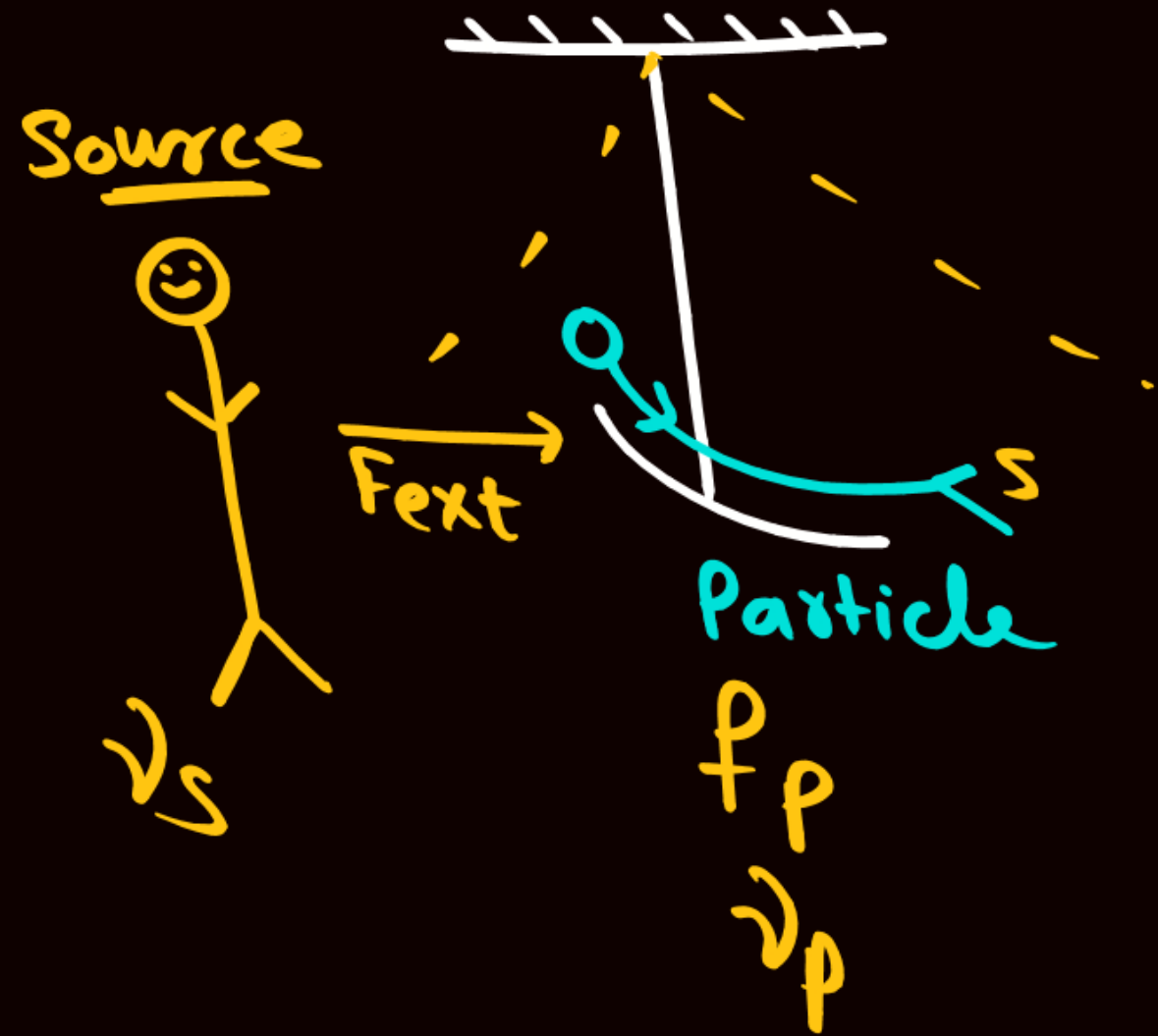
ν_3

Resonance

The condition in which the frequency of Source is equal to the frequency of Particle.

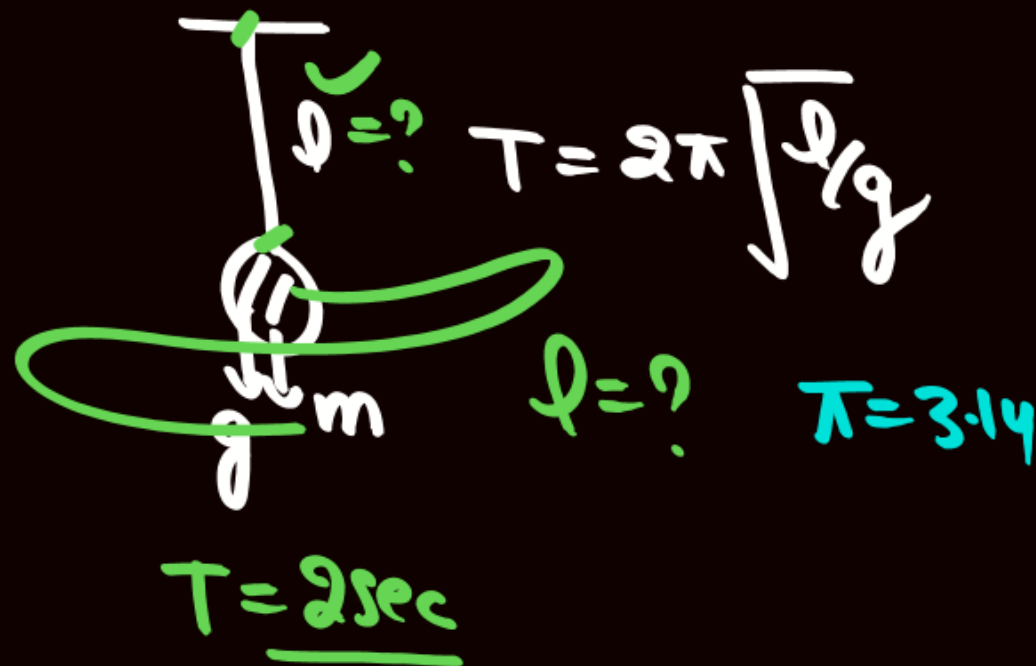
$$f_p = f_s$$

$$\lambda_p = \lambda_s$$



Second pendulum

The pendulum whose
time period = 2 sec



$$T = 2\pi \sqrt{l/g}$$

$$2 = 2\pi \sqrt{l/g}$$

$$\frac{2}{2\pi} = \sqrt{l/g}$$

$$\frac{1}{\pi} = \sqrt{l/g}$$

$$\frac{1}{3.14} = \sqrt{l/g}$$

$$\left(\frac{1}{3.14}\right)^2 = \left(\sqrt{l/g}\right)^2$$

$$\frac{1^2}{(3.14)^2} = l/g$$

$$\frac{1}{(3.14)^2} = \frac{l}{9.8} \quad g = 9.8$$

$$\frac{9.8}{9.8} = l$$

$$l = 1m$$

Periodic Function

$\sin \theta$



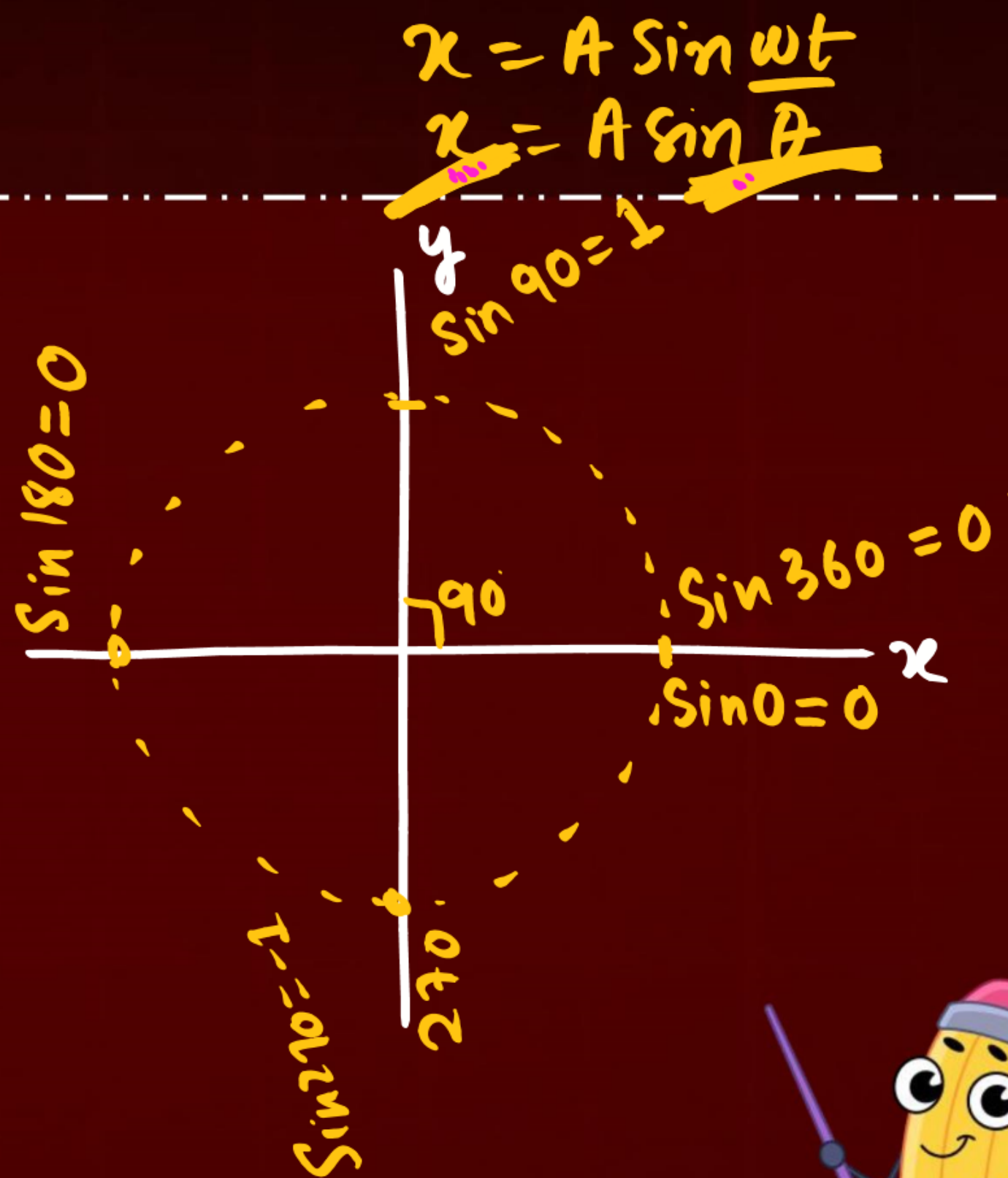
(*) The function which

Repeat itself

in Regular **Interval**

time

θ	✓ 0	90	✓ 180	270	✓ 360
Sin	0	1	0	-1	0
cos	1	0	-1	0	1



SUMMARY

$V \rightarrow \mathcal{X}$

D

Un-D

F

F

$\mathcal{V}$

R

P-Function

HOMework



↓↓
Copy → acche Sa
Bana

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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